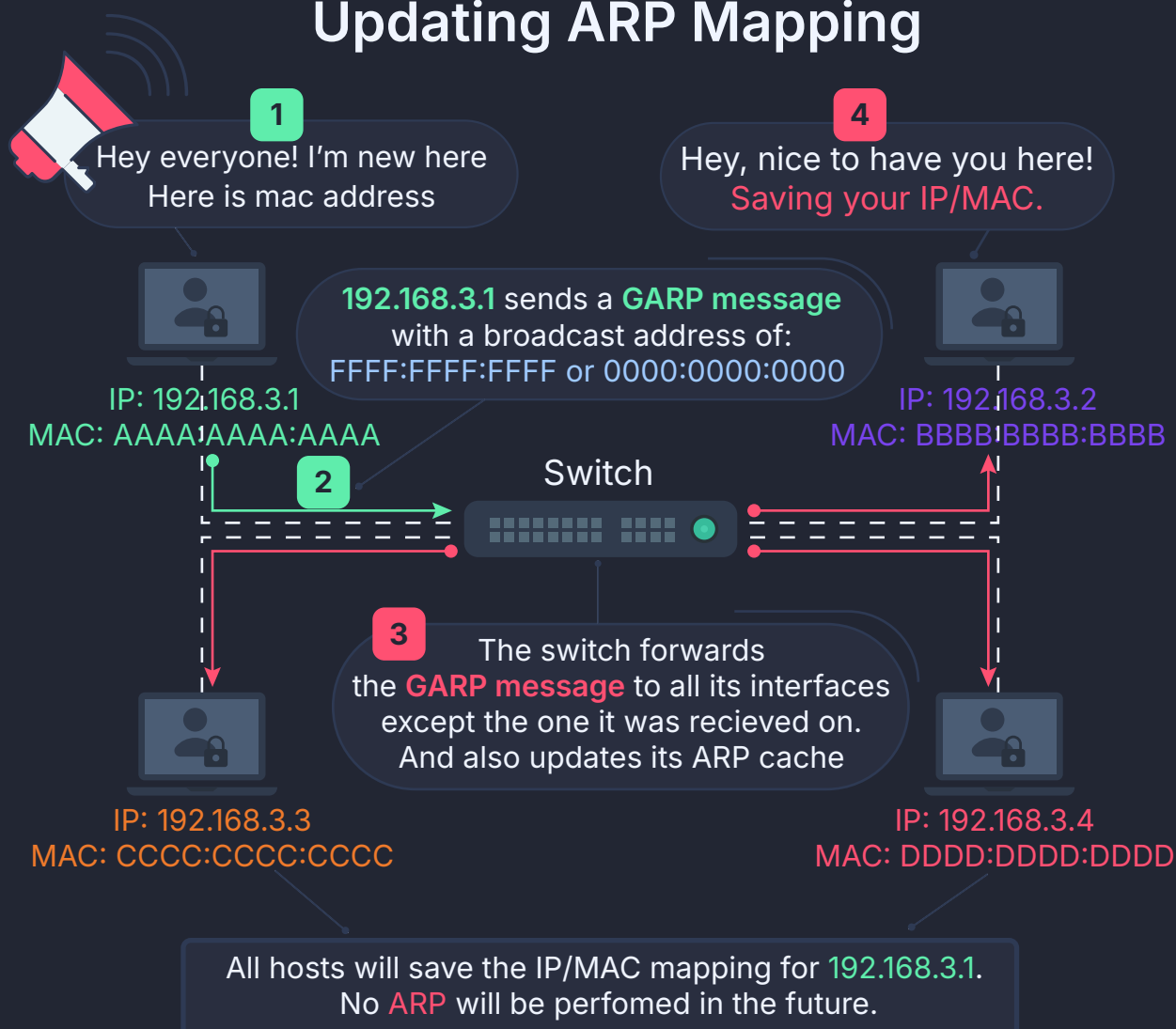


Updating ARP Mapping



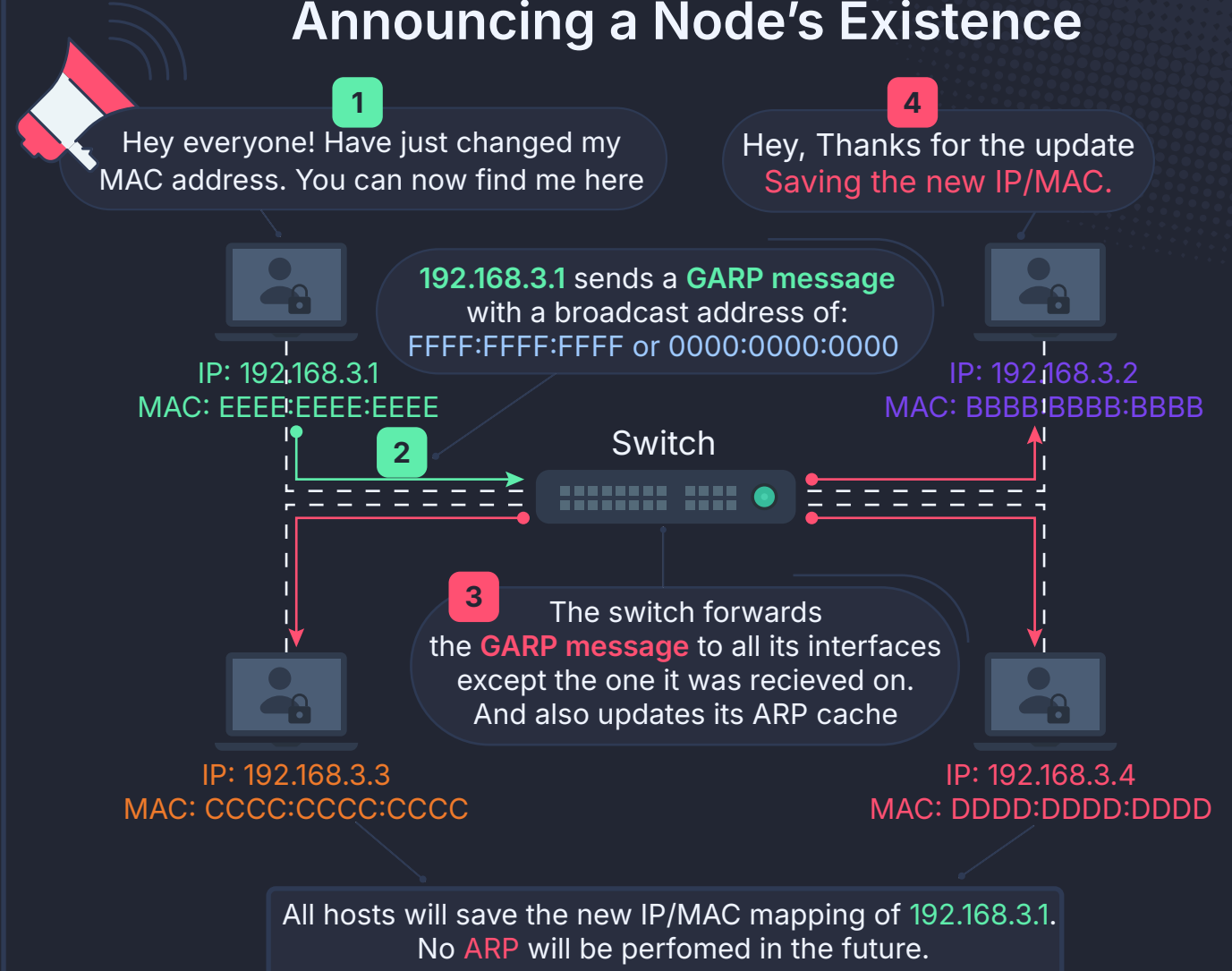
BEFORE 192.168.3.1 JOINS

MAC ADDRESS TABLE	
IP Address	MAC Address
192.168.3.2	BBBB:BBBB:BBBB
192.168.3.3	CCCC:CCCC:CCCC
192.168.3.4	DDDD:DDDD:DDDD

AFTER 192.168.3.1 JOINS

MAC ADDRESS TABLE	
IP Address	MAC Address
192.168.3.2	BBBB:BBBB:BBBB
192.168.3.3	CCCC:CCCC:CCCC
192.168.3.4	DDDD:DDDD:DDDD
192.168.3.1	AAAA:AAAA:AAAA

Announcing a Node's Existence



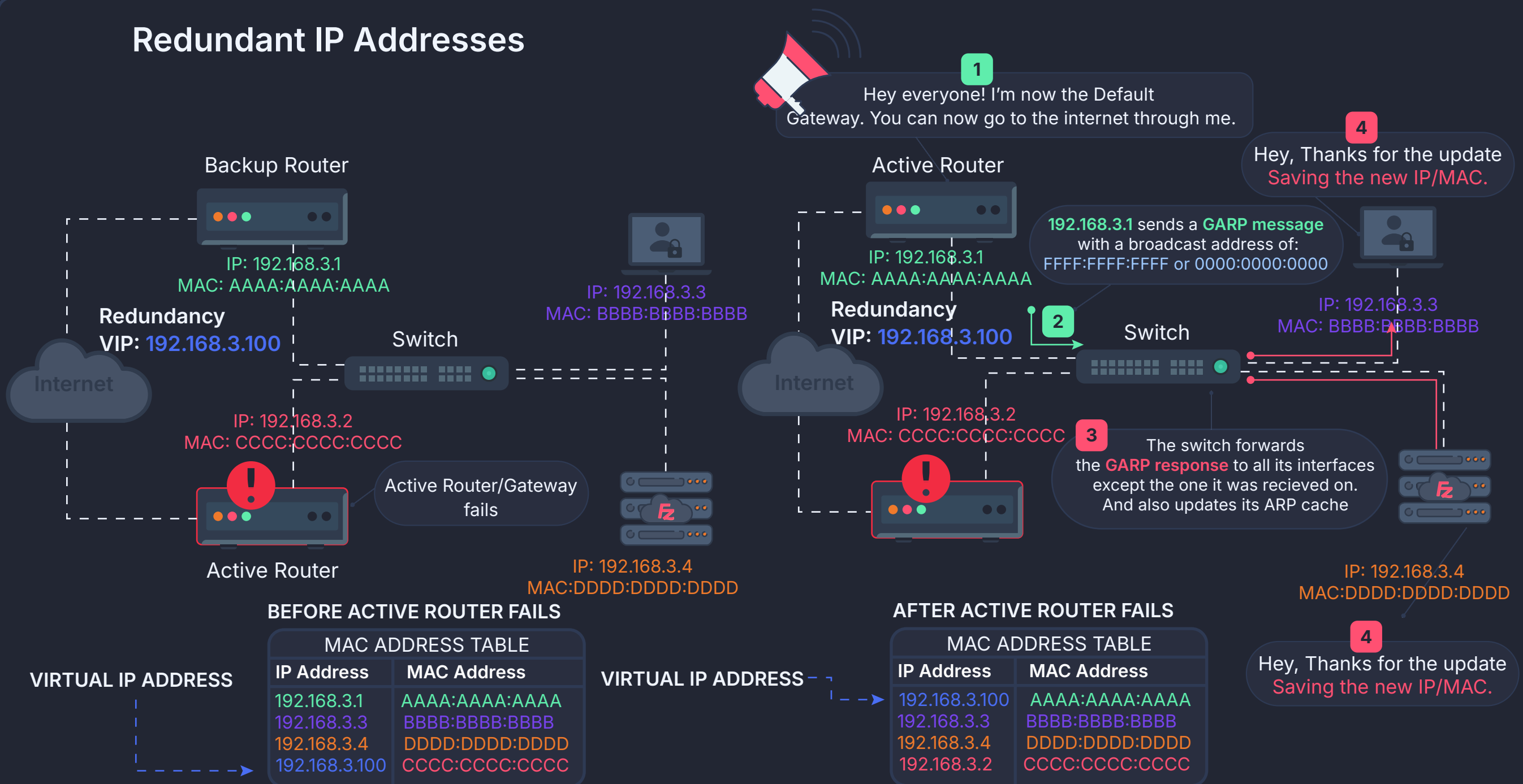
BEFORE MAC ADDR CHANGE

MAC ADDRESS TABLE	
IP Address	MAC Address
192.168.3.2	BBBB:BBBB:BBBB
192.168.3.3	CCCC:CCCC:CCCC
192.168.3.4	DDDD:DDDD:DDDD
192.168.3.1	AAAA:AAAA:AAAA

AFTER MAC ADDR CHANGE

MAC ADDRESS TABLE	
IP Address	MAC Address
192.168.3.2	BBBB:BBBB:BBBB
192.168.3.3	CCCC:CCCC:CCCC
192.168.3.4	DDDD:DDDD:DDDD
192.168.3.1	EEEE:EEEE:EEEE

Redundant IP Addresses



Gratuitous ARP (GARP) is an **ARP response** that is not prompted by an **ARP request**. Instead, it's initiated by a host to announce its presence or update its MAC address information across the network. **GARP** is commonly used in **HSRP (Hot Standby Router Protocol)** for failover, in **high availability (HA)** systems, and when a host changes its MAC address or a new host joins the network. It broadcasts this information to all devices, ensuring **ARP tables** are updated. Unlike regular **ARP responses**, which are **unicast**, **GARP messages** are sent via **broadcast**.